Heat Equation – 3D

The heat equation is a partial differential equation that describes how thermal energy spreads through a given material over time.

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

$$\alpha \ = \ \text{Thermal Diffusivity}$$

The partial differential equation becomes a stencil problem and similar methods are used to solve fluid and thermodynamic problems especially when related to partial differential equations. Array elements are updated according to some fixed pattern called a stencil. This results in following important rules

1. Cannot make outer time step loop parallel due to dependency on last step.
2. Cannot overwrite past state during operation. (Need double buffers)